# Unit 3 Part 2 Exam

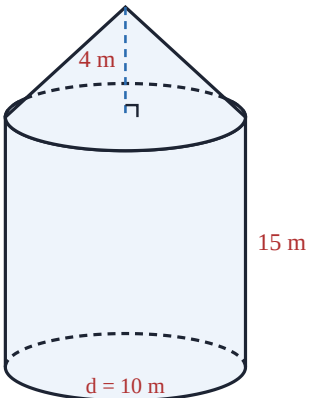
Math 9E  Part 2 file · Surface area + volume  6 questions · 25 marks  Name: _____

The most important questions predicted from this worksheet file. Show all your work. Answers are on the last page.

## Question 1

[5 marks]

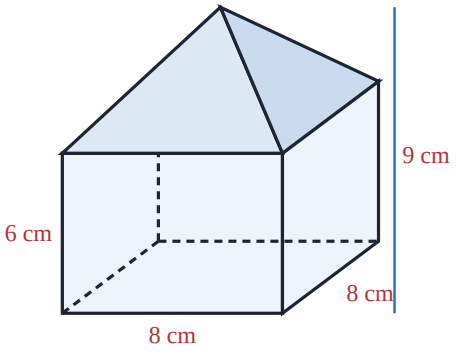
A silo is a cylinder with a conical roof. Diameter 10 m, cylinder height 15 m, roof height 4 m. The walls and roof need sealant (not the base). One bucket covers 30 $m^2$ and costs \$65. Find the area (2 decimal places), number of buckets and total cost.




## Question 2

[4 marks]

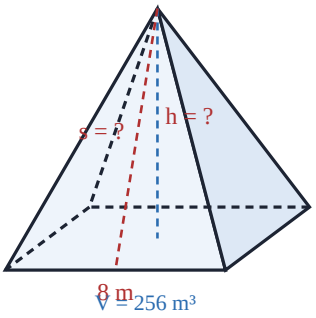
A square pyramid sits on top of a box with an 8 cm × 8 cm base and walls 6 cm tall. The total height of the shape is 9 cm. Find the total surface area.



## Question 3

[4 marks]

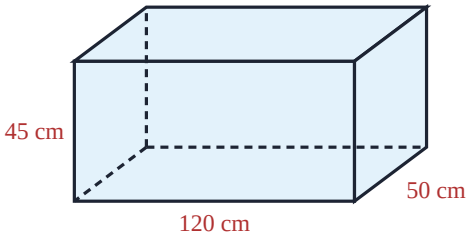
A right square pyramid has volume 256 m³ and base side 8 m. Find its height and its slant height to the nearest hundredth.




## Question 4

[3 marks]

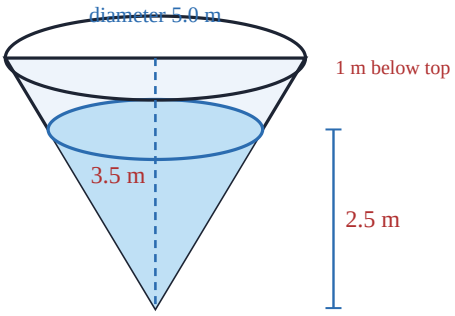
A fish tank is 120 cm × 50 cm × 45 cm. Fancy goldfish need at least 75 US gallons (1 gal = 3.785 L). (a) Is the tank big enough? (b) Keeping length and width the same, what height would hold exactly 75 gallons?



## Question 5

[5 marks]

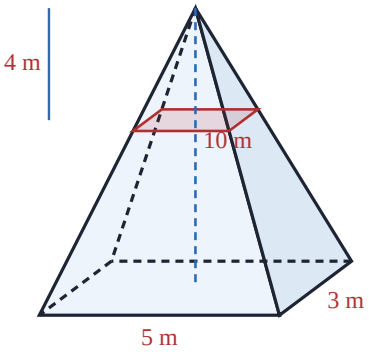
An underground tank is a right cone with its apex pointing down. Base diameter 5.0 m, height 3.5 m. (a) Find its capacity in kL to the nearest tenth. (b) How much water is in it when the water level is 1 m below the top? (c) What percentage full is it?




## Question 6

[4 marks]

A right rectangular pyramid has base 5 m × 3 m and height 10 m. A horizontal cut is made 4 m from the apex and the top pyramid is removed. Find the volume of the remaining piece.



## ANSWER KEY

**Q1** r = 5. Roof slant = $\sqrt{(5^2 + 4^2)} = \sqrt{41} \approx 6.403$. Walls $2\pi(5)(15) = 150\pi$; roof $\pi(5)(6.403) \approx 32.02\pi$. Total ≈ **571.82 m²**. 571.82 ÷ 30 = 19.06 ⇒ **20 buckets ⇒ $1300**.

**Q2** Pyramid height = 9 − 6 = 3. Slant = $\sqrt{(3^2 + 4^2)} = 5$. Walls 4(8)(6) = 192; base 64; triangles 4 · ½ · 8 · 5 = 80. Total = **336 cm²**.

**Q3** ⅓(64)h = 256 ⇒ **h = 12 m**. Slant = $\sqrt{(12^2 + 4^2)} = \sqrt{160} \approx$ **12.65 m**.

**Q4** (a) 120 × 50 × 45 = 270 000 cm³ = 270 L ÷ 3.785 ≈ 71.3 gal. **No, it is not big enough.** (b) 75 × 3.785 = 283.875 L = 283 875 cm³. h = 283 875 ÷ (120 × 50) ≈ **47.3 cm**.

**Q5** (a) V = ⅓π(2.5²)(3.5) ≈ 22.91 m³ ≈ **22.9 kL**. (b) Water depth = 2.5 m ⇒ k = 2.5/3.5 = 5/7. V = 22.91 × $(5/7)^3$ ≈ **8.3 kL**. (c) $(5/7)^3$ ≈ **36.4%**. Being only 1 m below the top still leaves the tank well under half full!

**Q6** $V_{big}$ = ⅓(15)(10) = 50 m³. k = 4/10 = 0.4 ⇒ $V_{small}$ = 50 × 0.064 = 3.2 m³. Remaining = **46.8 m³**.

## WHERE EACH QUESTION COMES FROM

| | |
|---|---|
| Q1 | Practice 3.2 #8 ★★★ (silo sealant) |
| Q2 | Practice 3.2 #11 (pyramid on a box) |
| Q3 | Practice 3.3 #5 ★★★ (pyramid slant from volume) |
| Q4 | Practice 3.3 #6 (fish tank in gallons) |
| Q5 | Practice 3.3 #4b ★★★+ (tank 1 m below top) |
| Q6 | Practice 3.3 #9 ★★★+ (cut pyramid) |